

TABLE VI
ESTIMATED SPECTRUM REQUIREMENTS FOR WIRELESS BROADBAND IN RURAL INDIA

Parameter	Year 2014
Number of blocks	6000
Average number of villages per block	100
Number of households per village	334
Percentage of households to be provided with broadband	12%
Number of households to be provided with broadband	40.1
Broadband rate per household (Mbps)	2
Broadband rate per community service centre	20
Bandwidth required per village (Mbps)	100.16
Bandwidth required per block (Mbps)	10016
Contention ratio	10:1
Maximum download data throughput required (Mbps)	1002
Download/upload traffic channel bandwidth ratio	2:1
Total data throughput required (Mbps)	1502.4
Estimated throughput per cell with 20MHz spectrum	90
Calculated total BTSs required to meet the throughput demand in a block	17

Source: TRAI recommendations on National Broadband Plan 2010

rural Indians stated not being aware of the Internet as the main reason for not using it.

Plans to take broadband to rural India

In August 2010, the Indian government prepared a white paper highlighting the importance of democratising information, especially government information, as lack of timely access to information leads to imbalance in power equations and, hence, exploitation of information have-nots. It suggested that government services such as birth and death certificates, land records, police reports, court papers and the like should be provided to the 250,000 *gram panchayats* using broadband connectivity via a reliable fibre-optic network. To achieve this, the report recommended the creation of a special-purpose vehicle, funded by USOF, to lay a fibre network till *gram panchayat* level.

Bandwidth estimate provided by TRAI in December 2010 shows a minimum of 2Mbps to 4Mbps for acceptable quality of service for common applications like email and

Internet browsing. Other applications such as telemedicine and multimedia based distance learning requiring higher bandwidths are to be provided selectively. Table VI shows the estimated spectrum requirements for year 2014.

In 2012, the Indian government formulated National Telecom Policy (NTP) 2012 that envisaged an "empowered and inclusive knowledge based society using telecommunications as a platform." The policy stated that the country has to achieve Broadband on Demand. A key objective was to reposition the mobile phone as a multifunction gadget so that it can serve as proof of identity, Internet banking device, training device and so on.

The 12th Five Year Plan (2012-2017) identifies the challenges in realising the vision of Broadband on Demand for rural and remote areas. To introduce 3G/4G/LTE/BWA services, it said that high-speed network infrastructure is essential and, hence, government intervention is required to provide incentives for roll-out in rural areas, investment in technology to bring down prices

of smartphones, investment in R&D for efficient spectrum utilisation and, most importantly, periodically review spectrum allocation to release unutilised spectrum.

Based on the above, a nationwide network, called National Optical Fibre Network (NOFN), funded by USOF, was planned to provide high-speed broadband to all *gram panchayats*. A special purpose vehicle, Bharat Broadband Network Ltd (BBNL), was created to set up and operate the network. Bandwidth will be provided to all eligible service providers in a non-discriminatory manner to deliver important services such as e-skilling and e-health.

BBNL plans to use existing fibre connectivity of three government-owned units, namely, BSNL, Railtel and Power Grid, for providing backhaul and laying the remaining cabling. BBNL has completed pilot projects covering 58 *gram panchayats* in three states, namely, Rajasthan, Andhra Pradesh and Tripura. Only 5000 *gram panchayats* have been connected. The original 2013 deadline for connecting all *gram panchayats* was revised to March 2017 and later advanced to December 2016.

In the meanwhile, a committee was formed to review NOFN project and provide recommendations. It gave its recommendations in March 2015. It has identified the causes for delay of NOFN project, some key reasons being linear architecture followed in laying cables, which will not provide acceptable reliability as expected by service providers, lack of alternatives in cases of difficult and hostile terrain, deficiencies in project management and inadequate planning with regards to how the network will be utilised.

The committee has extended the scope of NOFN keeping the objectives of Digital India in mind and re-named the project BharatNet. Apart from connecting *gram panchayats*,